22. Einstein Ring

A transparent unit sphere has a small hemispherical indent in its surface, located at $\hat{z}$. UV light shines on the indent from the $\hat{z}$ direction. The sphere is doped with fluorescent dye that has a chance to absorb UV light and reemit it as blue light in a random direction. The concentration of dye is small, so a UV photon typically travels a very long distance before being absorbed. Assume that light is either totally reflected or transmitted at interfaces.

A camera far away from the sphere in the $\hat{z}$ direction takes the following image; the center and boundary of the sphere are marked. If the sphere's indices of refraction for blue and UV light are n_b and n_u respectively:

(a) Find n_b. [6.0 pts]

(b) Find n_u. [5.0 pts]

(c) For which values (n_b, n_u) does a similar ring appear? [4.0 pts]

